

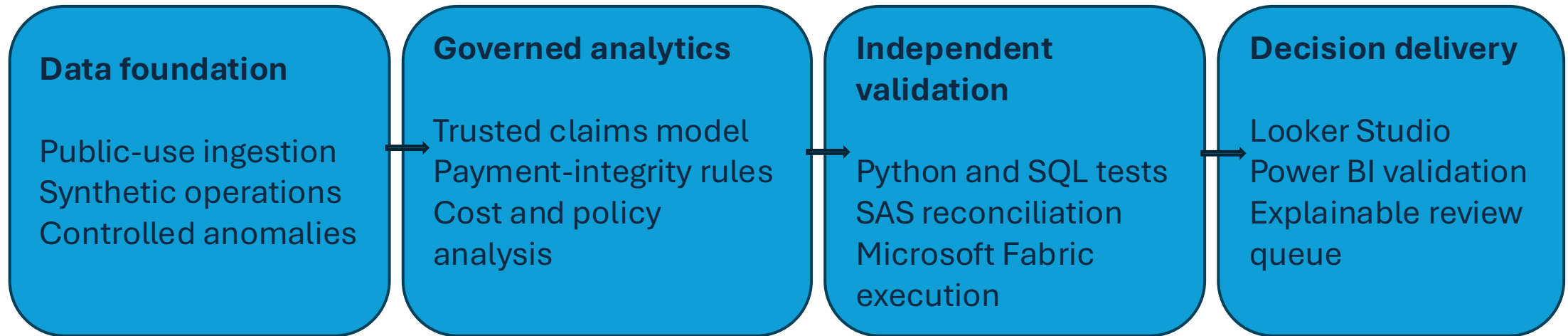
Healthcare Claims Payment Integrity & Cost Intelligence Platform

End-to-end analytics engineering with Python, SQL, SAS,
Microsoft Fabric, Looker Studio, and Power BI

Three questions drive the platform

- Why did healthcare claims costs change?
- Which payments merit explainable human review?
- Did the simulated claims-review policy change outcomes?
- How can every result remain reproducible, governed, and privacy-safe?

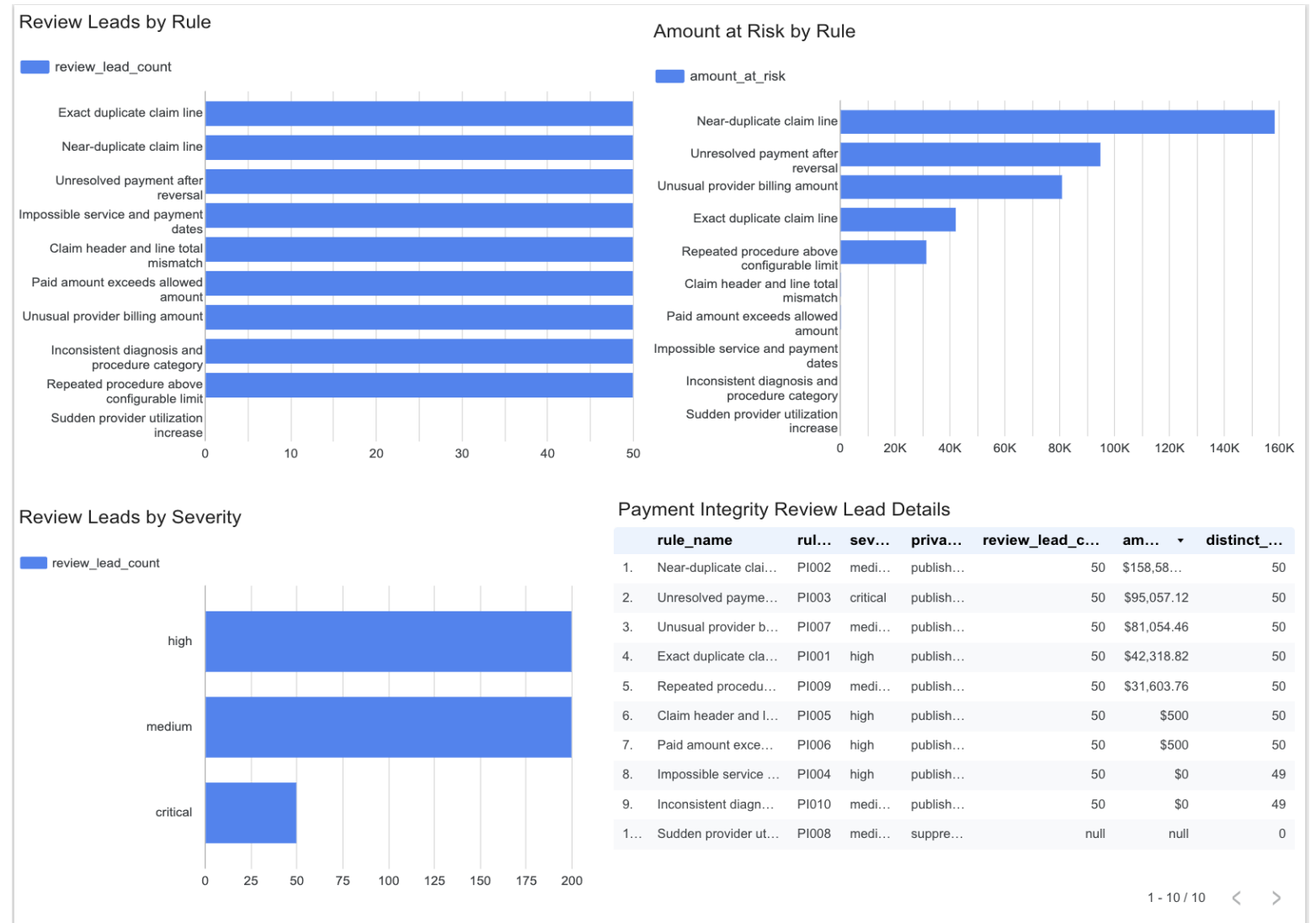
A governed path from claims data to decisions



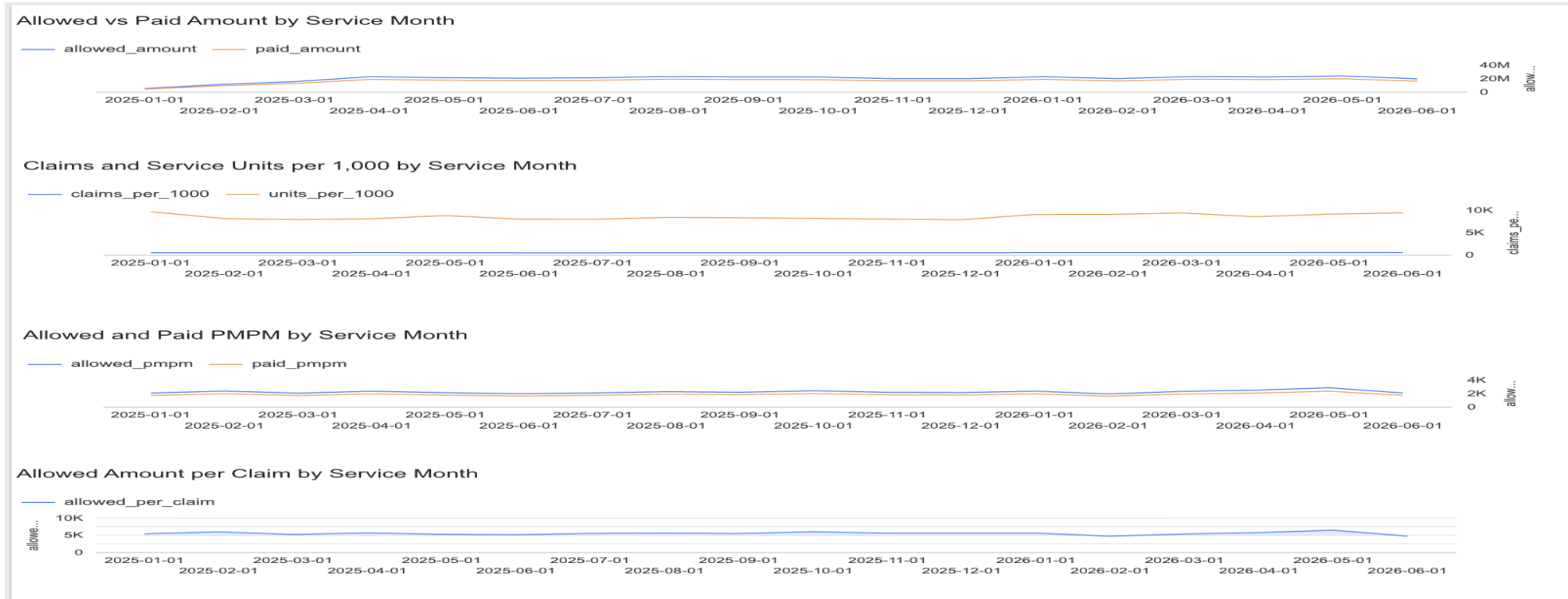
Versioned contracts • Deterministic builds • Privacy controls • Ground-truth isolation

Explainable review prioritization with strong test performance

- 10 explainable rules
- 500 controlled labels
- 88% precision
- 88% recall
- 0.01% false-positive rate
- 99.46% exposure recall
- \$14.83M detected expected exposure



Separate care utilization from payment cash flow



PMPM and per-1,000 rates • Concentration • Price-utilization-mix decomposition • Early-warning signals

20 quality checks passed • Cost decomposition reconciles within \$0.01

A statistically disciplined result can be “no demonstrated effect”

- 3,600 provider-month observations
- 200 providers
- 5 evaluated outcomes
- No statistically significant primary estimate
- All pre-trend tests passed
- All placebo tests passed
- Exposure weighting affected some estimates

Policy Effect Estimates

	outcome_na...	specification	coefficient	standard_error	confidence_interval_l...	confidence_inte...	p_value	observation_c...	provider_co...
1.	allowed_pmpm	balanced_panel	188.74	308.32	-415.56	793.04	0.54	3,575	200
2.	allowed_pmpm	unweighted_provider_mo...	-543.05	531.72	-1,585.22	499.11	0.31	3,575	200
3.	allowed_pmpm	winsorized_99_percent	220.67	304.92	-376.98	818.31	0.47	3,575	200
4.	claims_per_1000	balanced_panel	8.56	16.75	-24.27	41.39	0.61	3,575	200
5.	claims_per_1000	unweighted_provider_mo...	57.91	62.57	-64.72	180.55	0.35	3,575	200
6.	claims_per_1000	winsorized_99_percent	8.82	16.55	-23.61	41.25	0.59	3,575	200
7.	denial_rate	balanced_panel	-0.01	0.01	-0.02	0	0.25	3,599	200
8.	denial_rate	unweighted_provider_mo...	-0.01	0.01	-0.02	0	0.15	3,599	200
9.	denial_rate	winsorized_99_percent	-0.01	0.01	-0.02	0.01	0.29	3,599	200
10.	paid_pmpm	balanced_panel	160.25	255.57	-340.68	661.17	0.53	3,575	200
11.	paid_pmpm	unweighted_provider_mo...	-425.23	434.23	-1,276.32	425.86	0.33	3,575	200
12.	paid_pmpm	winsorized_99_percent	186.73	252.84	-308.84	682.3	0.46	3,575	200
13.	review_rate	balanced_panel	0.00	0	-0.01	0.01	0.81	3,599	200
14.	review_rate	unweighted_provider_mo...	0.01	0	-0	0.01	0.08	3,599	200
15.	review_rate	winsorized_99_percent	0.00	0	-0.01	0.01	0.86	3,599	200

Independent validation from local analytics to cloud execution

SAS reconciliation

181 comparisons passed
0 failures
0 missing SAS values

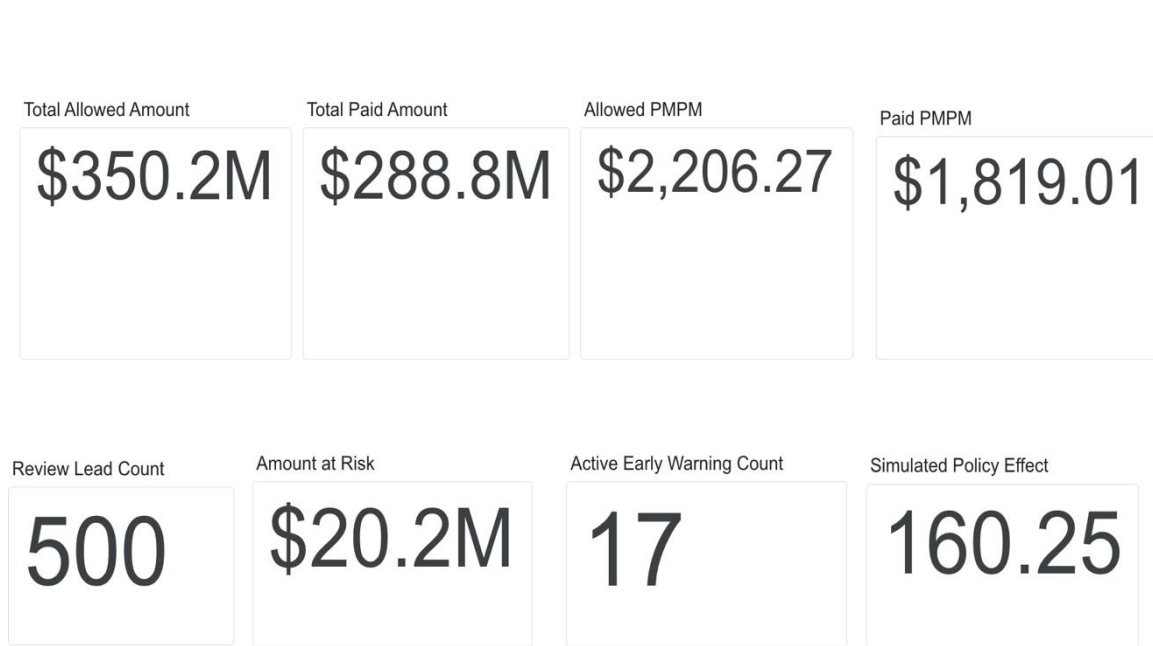
Independent reproduction of
governed analytical references

Microsoft Fabric

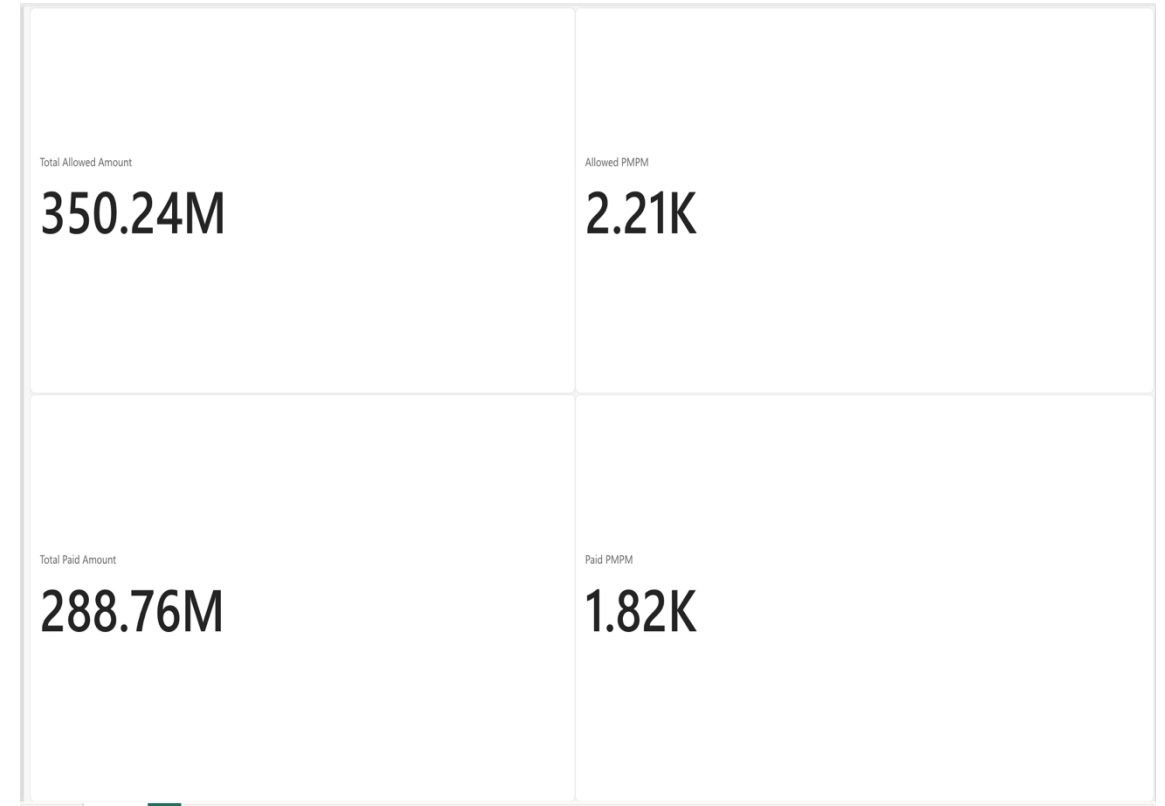
26 governed tables
23 ordinary • 3 restricted
33 reconciliation checks passed
Notebook and pipeline
succeeded
\$0 paid cloud-resource cost

Evaluation ground truth remained isolated from ordinary analytical features

One governed metric layer, validated across two BI tools



Looker Studio • Six-page primary dashboard



Power BI • Independent four-KPI validation

7 extracts • 14 governed KPIs • Exact count matches • Financial differences $\leq$ \$0.01 • 11-member privacy threshold

What this project demonstrates

- Analytics engineering and dimensional claims modeling
- Explainable payment-integrity review rules
- Cost intelligence and statistical policy evaluation
- Python, SQL, SAS, and Microsoft Fabric validation
- Looker Studio and Power BI delivery
- Automated testing, privacy governance, and reproducible evidence

A reproducible platform that explains cost changes, prioritizes review leads, evaluates policy effects honestly, and delivers governed results across multiple tools.